

JPA

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Till now we have made many project in core java but the major problem in when I restart my system all data which I store is lost.

It means we are not having permanent storage of data

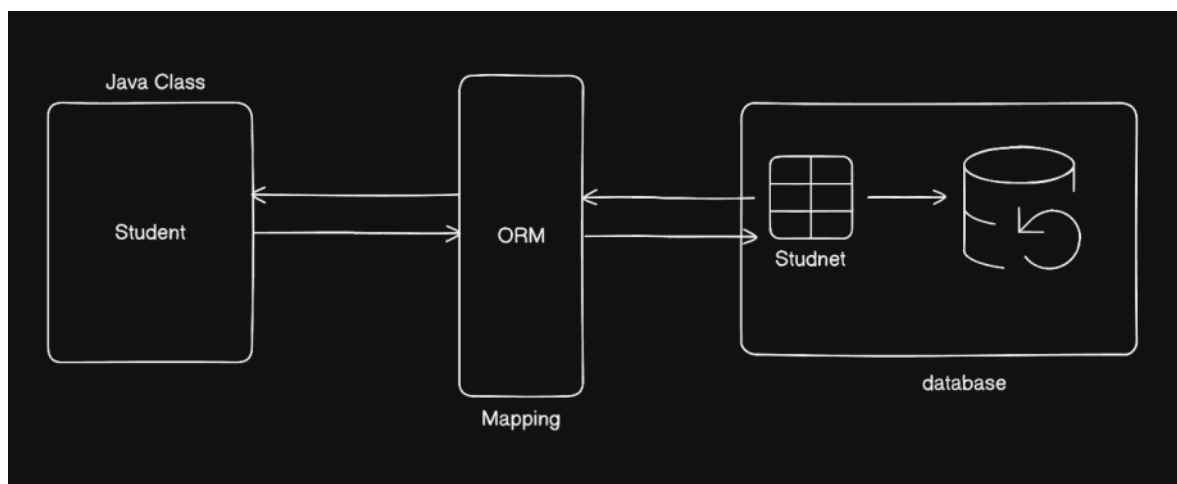
Then how we solve this problem ?

We can use mysql, oracle databases to store data

But the another issue is here we can save data in MYSQL through sql queries with our current knowledge. Alos we know JDBC but working with JDBC is also a big task.

The what ?

ORM=> object relational mapping



=> Application contains several layers

1. Presentation layer
2. Service
3. Persistence layer

Spring data jpa is used to develop persistence layer of application

Spring data jpa providing readymade method to perform CRUD operation in DB tables

Spring data JPA providing readymade methods using two interfaces like below

1. CrudRepository(I)
2. JpaRepository(I)

Note: JpaRepository=CrudRepository + pagination methods + sorting methods

Spring data JPA Terminology

1. Data source object : it represent database connections

Note: Data source properties we can configure in application.properties file

2. Entity class: The class which is mapped with database table

@Entity
@Table
@Id
@Column

3. Repository interface : for every table we will create one repository interface to perform crud operation

```
Public interface StudentRepo extends CrudRepository<Student,Integer>
{
}
}
```

Note: for out repository interface, implementation will be provided in runtime by JVM.

Note: by using StudentRepo we can perform CRUD operation in STUDENT_TBL

4. Repository methods:
- a. Save(entity)
 - b. saveAll(Iterable<Entity> i)

Note: Above two methods are called as "UPSERT" method (UPDATE + INSERT)

- c. findById(Id)
- d. findAllById(List)
- e. findAll()
- f. Count()
- g. deleteById()
- h. deleteAll()

5. ORM properties: to automate some config
- a. Auto_ddl: dynamic schema generation
 - b. Show_sql: display generated queries

Project 2:

- 1. 2 table
- 2. Relationship

Department table

Id	Name	Location
1	IT	Noida
2	HR	Pune

Employee Table

Id	Name	Email	Salary	Department_id
1	Abhay	abhay@gmail.com	10000	1
2	Rahul	rahul@gmail.com	20000	1
3	Amit	amit@gmail.com	30000	2

Form employee point of view : m-->1

From dep point of view 1->m